Title of Your Project

Substitute ↑ this title ↑ with your project's title

3-cfu Project Work in Natural Language Processing

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DO NOT MODIFY THIS TEMPLATE - EXCEPT, OF COURSE FOR TITLE AND AUTHOR INFORMATION. IN THE FINAL VERSION, IN THE LaTeX SOURCE REMOVE THE `guidelines` OPTION FROM `\usepackage[guidelines]{nlpreport}`.

Abstract

The abstract is very brief summary of your report. Try to keep it no longer than 15-20 lines at most. Write your objective, your approach, and your main observations (what are the findings that make this report worthwhile reading?)

NOTICE: THIS REPORT'S LENGTH MUST NOT EXCEED **8 PAGES** PLUS AI REPORT, LINKS, REFERENCES AND APPENDICES.
THIS MEANS THAT YOU CANNOT FILL ALL SECTIONS TO MAXIMUM LENGTH. IT ALSO MEANS THAT, QUITE POSSIBLY, YOU WILL HAVE TO LEAVE OUT OF THE REPORT PART OF THE WORK YOU HAVE DONE OR OBSERVATIONS YOU HAVE. THIS IS NORMAL: THE REPORT SHOULD EMPHASIZE WHAT IS MOST SIGNIFICANT, NOTEWORTHY, AND REFER TO THE NOTEBOOK FOR ANYTHING ELSE.
FOR ANY OTHER ASPECT OF YOUR WORK THAT YOU WOULD LIKE TO EMPHASIZE BUT CANNOT EXPLAIN HERE FOR LACK OF SPACE, FEEL FREE TO ADD COMMENTS IN THE NOTEBOOK.
INTERESTING TEXT EXAMPLES THAT EXCEED THE MAXIMUM LENGTH OF THE REPORT CAN BE PLACED IN A DEDICATED APPENDIX AFTER THE REFERENCES.

1 Introduction

MAX 2 COLUMNS

The Introduction is an executive summary, which you can think of as an extended abstract. Start by writing a brief description of the problem you are tackling and why it is important.

Then give a short overview of known/standard-/possible approaches to that problem, if any, and what are their advantages/limitations.

After that, discuss your approach, and motivate why you follow that approach. If you are drawing inspiration from an existing model, study, paper, textbook example, challenge, . . . , be sure to add all the necessary references (Antici et al., 2021).[1]

Next, give a brief summary of your experimental setup: how many experiments did you run on which dataset. Last, make a list of the main results or take-home lessons from your work.

HERE AND EVERYWHERE ELSE: ALWAYS KEEP IN MIND THAT, CRUCIALLY, WHATEVER TEXT/CODE/-FIGURES/IDEAS/... YOU TAKE FROM ELSEWHERE MUST BE CLEARLY IDENTIFIED AND PROPERLY REFERENCED IN THE REPORT.

2 Background

MAX 2 COLUMNS

The Background section is where you briefly provide whatever background information on the domain or challenge you're addressing and/or on the techniques/approaches you're using, that (1) you think is necessary for the reader to understand your work and design choices, and (2) is not something that has been explained to you during the NLP course (to be clear: do NOT repeat explanations of things seen in class, we already know that stuff). If you adapt paragraphs from articles, books, online resources, etc: be sure to clarify which parts are yours and which ones aren't.

3 Data

MAX 2 COLUMNS

Provide a brief description of your data including some statistics and pointers (references to articles/URLs) to be used to obtain the data. Describe any pre-processing work you did. Links to datasets must be placed later in Section 9.

[1] Add only what is relevant.

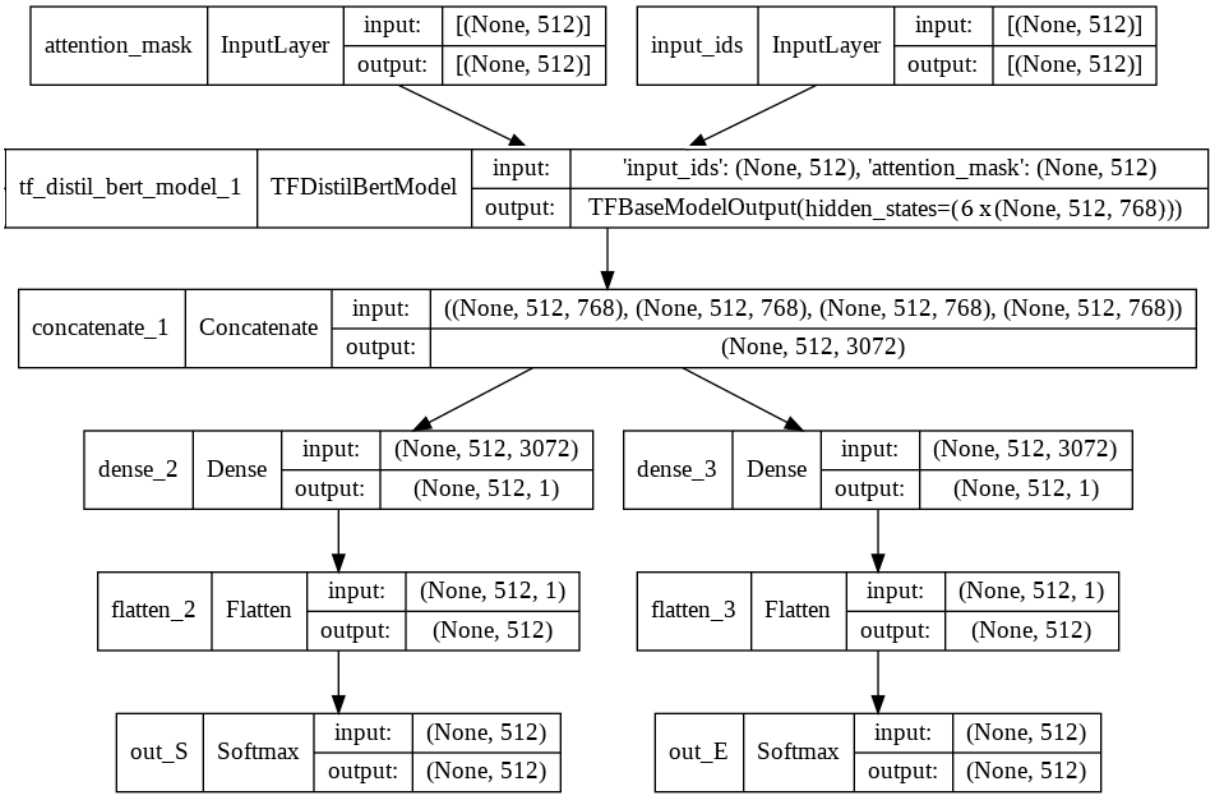


Figure 1: Model architecture

4 Approach

MAX 4 COLUMNS

Describe the system or systems you have implemented (architectures, pipelines, etc.), and used to run your experiments. If you reuse parts of code written by others, be sure to make very clear your original contribution in terms of

- architecture: is the architecture your design or did you take it from somewhere else
- coding: which parts of code are original or heavily adapted? adapted from existing sources? taken from external sources with minimal adaptations?

This also applies to AI-assisted coding.
It is a good idea to add figures to illustrate your pipeline and/or architecture(s) (see Figure 1)

5 Experiments

MAX 3 COLUMNS

Describe how you set up your experiments: which architectures/configurations you used, which hyper-parameters and what methods used to set them, which optimizers, metrics, etc.
Then, **use tables** to summarize your findings (numerical results) in validation and test. If you don't have experience with tables in LaTeX, you might want to use LaTeXtable generator to quickly create a table template.

6 Discussion

MAX 3 COLUMNS

Here you should make your analysis of the results you obtained in your experiments. Your discussion should be structured in two parts:

- discussion of quantitative results (based on the metrics you have identified earlier; compare with baselines);
- error analysis: show some examples of odd-/wrong/unwanted outputs; reason about why you are getting those results, elaborate on what could/should be changed in future developments of this work.

You don't need to copy the examples here if they're too long to fit: you can refer to the notebook, or use the Appendix at the end of the report.

7 Conclusion

MAX 1 COLUMN

In one or two paragraphs, recap your work and main results. What did you observe? Did all go according to expectations? Was there anything surprising or worthwhile mentioning? After that, discuss the main limitations of the solution you have implemented, and indicate promising directions for future improvement.

8 AI Report

THIS SECTION IS MANDATORY AND DOES NOT CONTRIBUTE TO THE PAGE LIMIT

Report on your use of AI for carrying out this work. State clearly what part of this work relied on AI assistance. List the specific tools used and how you integrated the output into your own work.

References

Francesco Antici, Luca Bolognini, Matteo Antonio Inajetovic, Bogdan Ivasiuk, Andrea Galassi, and Federico Ruggeri. 2021. SubjectivITA: An Italian corpus for subjectivity detection in newspapers. In *CLEF*, volume 12880 of *Lecture Notes in Computer Science*, pages 40–52. Springer.

9 Links to external resources

THIS SECTION IS OPTIONAL AND DOES NOT CONTRIBUTE TO THE PAGE LIMIT

Insert here a link to your GitHub or any other public repo where one can find your code (only if you did not submit your code on Virtuale) or other software resources you have used. Do not insert code in this report.

10 Appendix

THIS SECTION IS OPTIONAL AND DOES NOT CONTRIBUTE TO THE PAGE LIMIT

Use the appendix if you need to show examples that do not fit in the two-page limit or anything else that you consider absolutely important but impossible to fit in the 2-page report or in the notebook.